

Accounting Business and Society

–Non-current Assets

WEEK 4

Content

- 01 Key concepts**
- 02 Assessment**
- 03 Tutorial Activity**

Key concepts

01 NCA— Recording—Expensing—Reporting

02 Accounts for Depreciation

03 Adjustments made during NCA' s life

04 Disposal of Non-current Assets

PPE & Intangibles Terminology

Asset Account (Balance Sheet)		Related Expense Account (Income Statement)
Property, Plant and Equipment		
	Freehold Land	None
	Leasehold land	Depreciation
	Buildings, Machinery & Equipment	Depreciation
	Furniture & Fixtures	Depreciation
	Land Improvements	Depreciation
	Natural Resources	Depletion
Intangibles		
	Intangibles with finite useful lives	Amortization
	Intangibles with indefinite useful lives	None

#1 Acquisition of Non-current Assets

**Sum of all costs
incurred to bring
the asset to its
intended use**

Acquisition of Non-current Assets

Working rule for measuring the cost of an asset:

The cost of any asset is the sum of all the costs incurred to bring the asset to its intended use.

- **Costs include:**

- Purchase price
- Taxes
- Commissions
- Other costs to make the asset ready for use

Asset	Costs included
Land	Purchase price, commissions, survey & legal fees, and back property taxes paid; grading and removing unwanted buildings
Land Improvements	Fencing, paving, security systems & lighting
Building – Constructed	Architectural fees, contractors' charges, materials, labor, and overhead; interest on funds borrowed
Building – Purchased	Purchase price, broker's commission, taxes paid and all costs to repair and renovate building
Equipment	Purchase price, transportation, insurance in transit, sale tax, installation and testing

Acquisition of Non-current Assets

FedEx signs a \$300,000 note payable to purchase 20 acres of land. FedEx also pays \$10,000 for real estate commission, \$8,000 of back property tax, \$5,000 for removal of an old building, a \$1,000 survey fee, and \$260,000 to pave the parking lot—all in cash. **What is FedEx's cost of this land?**

Purchase price of land		\$300,000
Add related costs:		
Real estate commission	\$10,000	
Back property tax.....	8,000	
Removal of building.....	5,000	
Survey fee.....	<u>1,000</u>	
Total related costs		<u>24,000</u>
Total cost of land		<u><u>\$324,000</u></u>

Acquisition of Non-current Assets

FedEx signs a \$300,000 note payable to purchase 20 acres of land. FedEx also pays \$10,000 for real estate commission, \$8,000 of back property tax, \$5,000 for removal of an old building, a \$1,000 survey fee, and \$260,000 to pave the parking lot—all in cash.

A1				
	A	B	C	D
1		Land	324,000	
2		Note Payable		300,000
3		Cash		24,000
4				

Acquisition of Non-current Assets

- **Lump-Sum (or Basket) Purchases of Assets**
 - Purchase several assets for a single lump-sum amount
 - Total cost divided among the assets based on relative market values
 - Technique is called the **relative-sales-value method**

Acquisition of Non-current Assets

FedEx purchases land and a building in Denver. The building sits on two acres of land, and the combined purchase price of land and building is \$2,800,000. An appraisal indicates that the land's market value is \$300,000 and that the building's market value is \$2,700,000.

Asset	Market (Sales) Value		Total Market Value		Percentage of Total Market Value		Total Cost		Cost of Each Asset
Land	\$ 300,000	÷	\$3,000,000	=	10%	×	\$2,800,000		\$ 280,000
Building	<u>2,700,000</u>	÷	3,000,000	=	<u>90%</u>	×	\$2,800,000		<u>2,520,000</u>
Total	<u><u>\$3,000,000</u></u>				<u><u>100%</u></u>				<u><u>\$2,800,000</u></u>

Acquisition of Non-current Assets

FedEx purchases land and a building in Denver. The building sits on two acres of land, and the combined purchase price of land and building is \$2,800,000. An appraisal indicates that the land's market value is \$300,000 and that the building's market value is \$2,700,000.

A1				
	A	B	C	D
1		Land	280,000	
2		Building	2,520,000	
3		Cash		2,800,000
4				

#2 Accounting for Depreciation

- **Depreciation**

- Process of allocating a plant asset's cost to expense over its life
- Necessary as plant assets wear out, grow obsolete, and lose value over time
- Allocates cost against revenue it helps earn each period
- Depreciation expense is reported on the income statement
- Land is not depreciated

Accounting for Depreciation



Estimated useful life, 20 years



Annual revenue generated, \$9 million

minus

Annual depreciation expense, \$1.6 million*

*\$32 million ÷ 20 years = \$1.6 million per year.

How to Measure Depreciation

- **Cost**
 - Purchase price and all costs to get plant asset ready for use, known amount
- **Estimated Useful Life**
 - Length of service **expected** from using the asset, estimated amount
- **Estimated Residual Value**
 - **Expected** cash value of an asset at the end of its useful life, estimated amount

Depreciation Methods (1 of 6)

- Three main depreciation methods:
 - Straight line
 - Units of production
 - Reducing Balance,
 - e.g. Double-declining balance

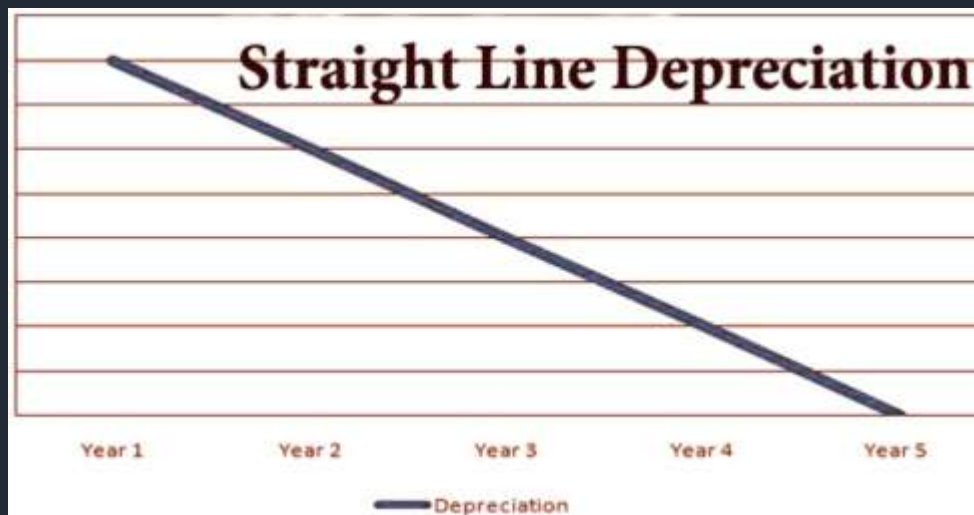
Depreciation Computation Data (1 of 4)

Data Item	Amount
Cost of truck	\$41,000
Less: Estimated residual value	<u>(1,000)</u>
Depreciable cost	<u><u>\$40,000</u></u>
Estimated useful life:	
Years	5 years
Units of production	100,000 units [miles]

Depreciation Methods (2 of 6)

- **Straight-Line Method**

- Equal amount of depreciation assigned each period
- Depreciable cost divided by useful life to determine annual depreciation expense



Depreciation Computation Data (2 of 4)

Data Item	Amount
Cost of truck	\$41,000
Less: Estimated residual value	<u>(1,000)</u>
Depreciable cost	<u><u>\$40,000</u></u>
Estimated useful life:	
Years	5 years
Units of production	100,000 units [miles]

$$\begin{aligned}\text{Straight-line depreciation per year} &= \frac{\text{Cost} - \text{Residual value}}{\text{Useful life, in years}} \\ &= \frac{\$41,000 - \$1,000}{5} \\ &= \mathbf{\$8,000}\end{aligned}$$

Depreciation Methods (3 of 6)

Straight-Line Method

A1				
	A	B	C	D
1		Depreciation Expense—Truck	8,000	
2		Accumulated Depreciation—Truck		8,000
3				

Straight-Line Depreciation Schedule for Truck

	A1						
	A	B	C	D	E	F	G
1	Date	Cost	Rate*	Depreciable Cost	Yearly Expense	Accum. Deprec.	Book Value
2	1/1/2013	41,000		40,000			41,000
3	12/31/2013		0.2	40,000	8,000	8,000	33,000
4	12/31/2014		0.2	40,000	8,000	16,000	25,000
5	12/31/2015		0.2	40,000	8,000	24,000	17,000
6	12/31/2016		0.2	40,000	8,000	32,000	9,000
7	12/31/2017		0.2	40,000	8,000	40,000	1,000
8							

*1/years of useful life = $1/5 = 0.2$

Depreciation Methods (4 of 6)

- **Units-of-Production Method**

- Fixed amount of depreciation is assigned to each unit of output produced by the asset
- Depreciable cost is divided by useful life—in units of production—to determine fixed amount per unit
- Per-unit amount then multiplied by number of units produced each period to compute depreciation expense

Depreciation Computation Data (3 of 4)

Units-of-production depreciation per unit of output

Data Item	Amount
Cost of truck	\$41,000
Less: Estimated residual value	<u>(1,000)</u>
Depreciable cost	<u><u>\$40,000</u></u>
Estimated useful life:	
Years	5 years
Units of production	100,000 units [miles]

$$\begin{aligned} &= \frac{\text{Cost} - \text{Residual value}}{\text{Useful life, in units of production}} \\ &= \frac{\$41,000 - \$1,000}{100,000 \text{ miles}} = \mathbf{\$0.40 \text{ per mile}} \end{aligned}$$

Exhibit 7-6 Units-of-Production (UOP) Depreciation Schedule for Truck

Assume that FedEx expects to drive the truck 20,000 miles during the first year, 30,000 during the second, 25,000 during the third, 15,000 during the fourth, and 10,000 during the fifth. Exhibit 7-6 shows the UOP depreciation schedule.

A1							
	A	B	C	D	E	F	G
1	Date	Cost	Rate per unit	Number Units	Yearly Expense	Accum. Deprec.	Book Value
2	1/1/2013	41,000					41,000
3	12/31/2013		0.4	20,000	8,000	8,000	33,000
4	12/31/2014		0.4	30,000	12,000	20,000	21,000
5	12/31/2015		0.4	25,000	10,000	30,000	11,000
6	12/31/2016		0.4	15,000	6,000	36,000	5,000
7	12/31/2017		0.4	10,000	4,000	40,000	1,000
8							

Depreciation Methods (5 of 6)

- **Double-Declining-Balance Method**

- Writes off a larger amount of cost near the start of asset's useful life
- Most frequently used accelerated depreciation method
- Multiplies asset's declining book value at the beginning of the year by a constant percentage, two times the straight-line depreciation rate

Depreciation Computation Data (4 of 4)

Data Item	Amount
Cost of truck	\$41,000
Less: Estimated residual value	<u>(1,000)</u>
Depreciable cost	<u><u>\$40,000</u></u>
Estimated useful life:	
Years	5 years
Units of production	100,000 units [miles]

$$\begin{aligned}\text{DDB depreciation rate per year} &= \frac{1}{\text{Useful life, in years}} \times 2 \\ &= \frac{1}{5 \text{ years}} \times 2 \\ &= 20\% \times 2 = \mathbf{40\%}\end{aligned}$$

Exhibit 7-7 Double-Declining-Balance (DDB) Depreciation Schedule for Truck

A1						
	A	B	C	D	E	F
1	Date	Cost	DDB Rate	Yearly Expense	Accum. Deprec.	Book Value
2	1/1/2013	41,000				41,000
3	12/31/2013		0.4	16,400	16,400	24,600
4	12/31/2014		0.4	9,840	26,240	14,760
5	12/31/2015		0.4	5,904	32,144	8,856
6	12/31/2016		0.4	3,542	35,686	5,314
7	12/31/2017		0.4	4,314	40,000	1,000*
8						

* Final-year depreciation is a plug amount needed to reduce asset book value to estimated salvage value

Depreciation Methods (6 of 6)

- **DDB method differs from others in two ways:**
 - Residual value is ignored initially
 - Final year depreciation is a “plug”

Comparing Depreciation Methods (1 of 2)

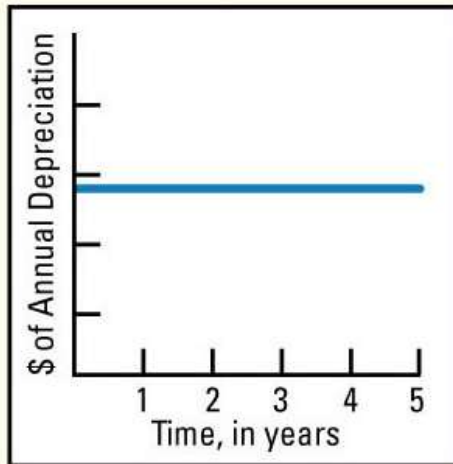
- **Straight-line**
 - Best for plant assets that generate revenue evenly over time
- **Units-of-production**
 - Best for assets that wear out because of use
- **Double-declining-balance**
 - Best for assets that generate more revenue earlier in their useful life

Comparing Depreciation Methods (2 of 2)

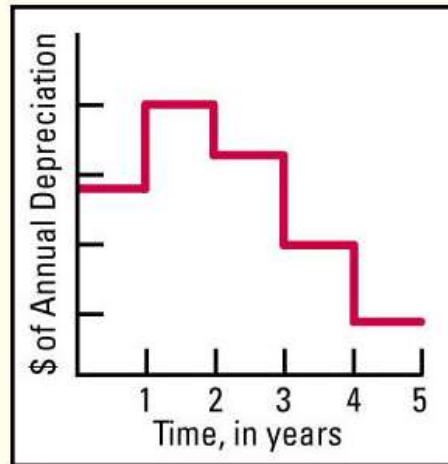
Year	<i>Amount of Depreciation per Year</i>		
	Straight-Line	Units-of-Production	Accelerated Method Double-Declining Balance
1	\$ 8,000	\$ 8,000	\$16,400
2	8,000	12,000	9,840
3	8,000	10,000	5,904
4	8,000	6,000	3,542
5	8,000	4,000	4,314
Total	<u>\$40,000</u>	<u>\$40,000</u>	<u>\$40,000</u>

Depreciation Patterns Through Time

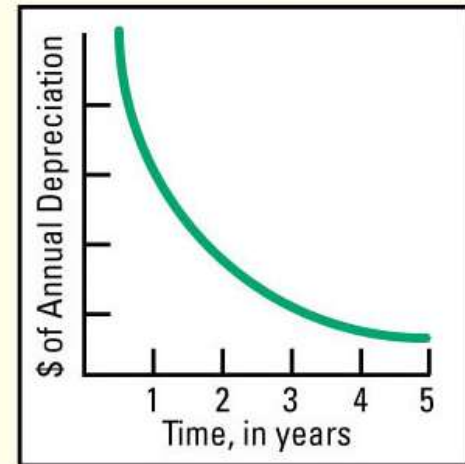
Straight-Line



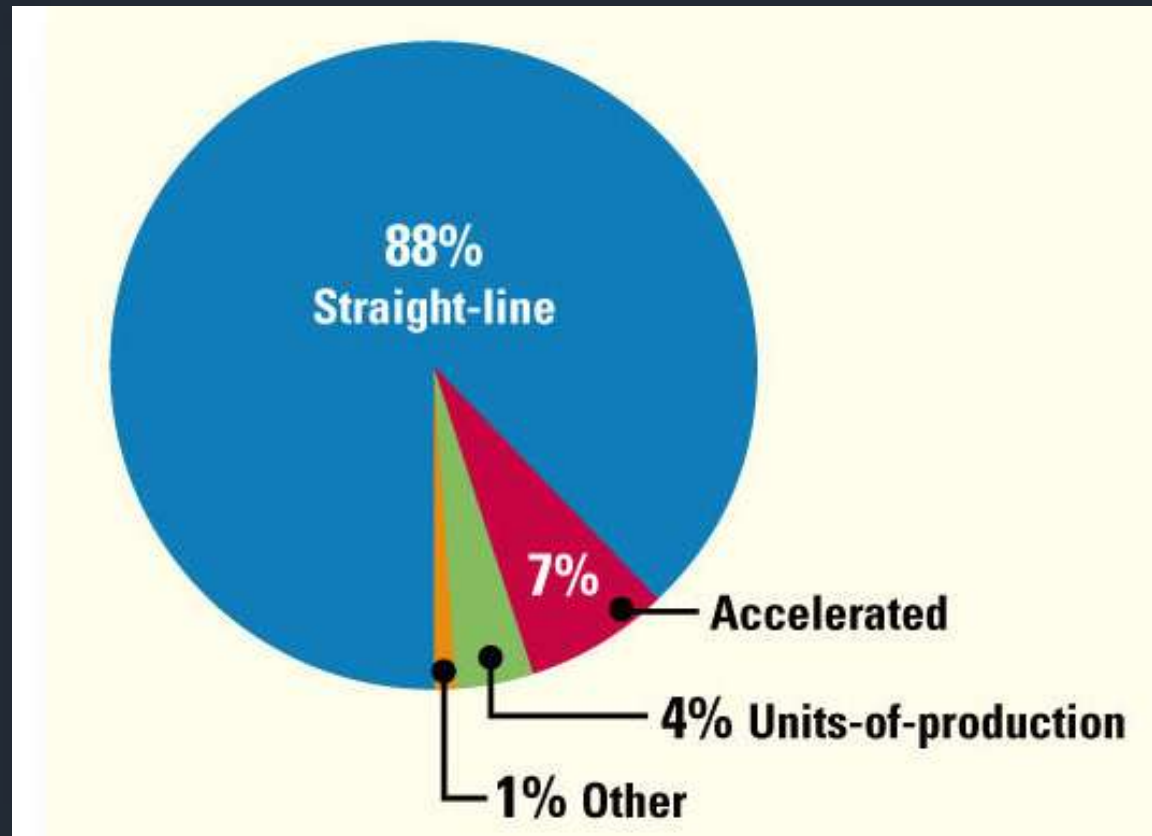
Units-of-Production



Accelerated (DDB)



Depreciation Methods Used by 600 Companies



Depreciation for Tax Purposes

- Most companies use straight-line depreciation for financial reporting, but the double-declining-balance method for tax purposes.
- Accelerated depreciation provides fastest tax deductions → Tax deductions reduce income taxes → Reduced income taxes helps conserve cash

The Cash Flow Advantage of Accelerated Depreciation for Tax Purposes

	SL	Accelerated
1 Cash revenue.....	\$400,000	\$400,000
2 Cash operating expenses	<u>300,000</u>	<u>300,000</u>
3 Net cash provided by operations before income tax.....	100,000	100,000
4 Depreciation expense (a noncash expense).....	<u>8,000</u>	<u>16,400</u>
5 Income before income tax	<u>\$ 92,000</u>	<u>\$ 83,600</u>
6 Income tax expense (30%)	<u>\$ 27,600</u>	<u>\$ 25,080</u>
Cash-flow analysis:		
7 Net cash provided by operations before tax	\$100,000	\$100,000
8 Income tax expense	<u>27,600</u>	<u>25,080</u>
9 Net cash provided by operations	<u>\$ 72,400</u>	<u>\$ 74,920</u>
10 Extra cash available for investment if DDB is used (\$74,920 – \$72,400).....		<u>\$ 2,520</u>

Modified Accelerated Cost Recovery System (MACRS)

Class Identified by Asset Life (years)	Representative Assets	Depreciation Method
3	Race horses	DDB
5	Automobiles, light trucks, computers	DDB
7	Office furniture, fixtures	DDB
10	Other equipment	DDB
15	Sewage-treatment plants	150% DB
20	Certain real estate	150% DB
27½	Residential rental property	SL
39	Nonresidential rental property	SL

Depreciation for Partial Years

Suppose FedEx purchases a warehouse building on September 1 for \$500,000. The building's estimated life is 20 years, and its estimated residual value is \$80,000. FedEx's accounting year ends on May 31.

$$\text{Full-year depreciation} \quad \frac{\$500,000 - \$80,000}{20} = \$21,000$$

$$\text{Partial year depreciation} \quad \$21,000 \times \frac{9}{12} = \$15,750$$

#3 Adjustments made during NCA' s life--change in estimates

- After an asset is in use, managers may change its useful life based on experience or new information.
 - Called a **change in accounting estimate**

Adjustments made during NCA's life—change in estimates

Assume a Disney hot dog stand cost \$50,000 and originally had a useful life of 10 years (no residual value). Disney used the asset for four years, resulting in accumulated depreciation of \$20,000 and a remaining book value of \$30,000. Management now believes the

A1				
	A	C	D	G
1		Depreciation Expense—Hot Dog Stand	3,000	
2		Accumulated Depreciation—Hot Dog Stand		3,000
3				

$$\frac{\text{Asset's remaining depreciable book value}}{\text{New estimated useful life}} = \text{New annual depreciation}$$

$$\frac{30,000}{10} = \$3,000$$

Changing Useful Life

Remaining depreciable book value

(New) Estimated remaining useful life

Adjustments made during NCA' s life--additional capital expenditures incurred for NCAs

- **Capital expenditure** → increase the asset's capacity or extends its useful life
- **Capitalized** means the cost is added to an asset account rather than being expensed immediately



Exhibit 7-2 Capital Expenditures vs. Immediate Expenses

Record an Asset for Capital Expenditures	Record Repair and Maintenance Expense (Not an Asset) for an Expense
Extraordinary repairs: Major engine overhaul Modification of body for new use of truck Addition to storage capacity of truck	Ordinary repairs: Repair of transmission or other mechanism Oil change, lubrication, and so on Replacement of tires and windshield, or a paint job

Impairment of PPE

- An asset is impaired when its carrying value is higher than its recoverable amount
 - Recoverable amount is the higher of fair value less cost to sell and value in use.

#4 Disposal of Non-Current Assets

- Before accounting for disposal, the business should bring the depreciation up to date to:
 - Measure the asset's final book value
 - Record the expense up to the date of disposal
- **Remove asset and related accumulated depreciation account from books**

Selling a Plant Asset

If cash received
is **greater** than
book value



GAIN

If cash received
is **less** than book
value



LOSS

Disposal of Non-Current Assets

Selling a Plant Asset. Suppose FedEx sells equipment on September 30, 2016 for \$7,300 cash. The equipment cost \$10,000 when purchased on January 1, 2013 and has been depreciated straight-line. It had an estimated life of 10 years with no residual value. The depreciation entry on September 30, 2016 is:

A1				
	A	C	D	G
1	Sep 30	Depreciation Expense ($\$10,000/10 \text{ years} \times 9/12$)	750	
2		Accumulated Depreciation—Equipment		750
3		<i>To update depreciation.</i>		
4				

Disposal of Non-Current Assets

Equipment	
Jan 1, 2013	10,000

Accumulated Depreciation—Equipment		
	Dec 31, 2013	1,000
	Dec 31, 2014	1,000
	Dec 31, 2015	1,000
	Sep 30, 2016	750
	Balance	3,750

= Book value
\$6,250

Cash received from sale of the asset		\$7,300
Book value of asset sold:		
Cost	\$10,000	
Less: Accumulated depreciation	<u>(3,750)</u>	<u>6,250</u>
Gain on sale of the asset.....		<u><u>\$1,050</u></u>

Disposal of Non-Current Assets

Selling a Plant Asset. Suppose FedEx sells equipment on September 30, 2016 for \$7,300 cash. The equipment cost \$10,000 when purchased on January 1, 2013 and has been depreciated straight-line. It had an estimated life of 10 years with no residual value. The entry to record the sale is:

A1				
	A	C	D	G
1	Sep 30	Cash	7,300	
2		Accumulated Depreciation—Equipment	3,750	
3		Equipment		10,000
4		Gain on Sale of Equipment		1,050
5		<i>To sell equipment.</i>		
6				

Disposal of Non-Current Assets

- **Exchanging a Plant Asset**
 - Trade in old assets for new ones
 - Nonmonetary exchanges
 - Based on fair values of assets involved
 - Differences between fair values of old and new assets results in a gain or loss

Disposal of Non-Current Assets

Exchanging a Plant Asset. Assume Papa John's Pizza's old delivery car cost \$9,000 with accumulated depreciation of \$8,000. Papa John's trades the old delivery car for a new one with a fair market value of \$15,000 and pays \$10,000 of cash.

A1				
	A	C	D	G
1		Delivery Auto (new)	15,000	
2		Accumulated Depreciation—Delivery Auto (old)	8,000	
3		Delivery Auto (old)		9,000
4		Cash		10,000
5		Gain on Exchange of Delivery Auto		4,000
6		<i>Traded in old delivery car for new auto.</i>		
7				

T-Accounts for Analyzing Plant Asset Transactions (1 of 2)

Building (or Equipment)	
Beg bal	
Cost of assets purchased	Cost of assets disposed of
End bal	

Accumulated Depreciation	
Accumulated depreciation of assets disposed of	Beg bal
	Depreciation expense for the current period
	End bal

Cash	
Cash proceeds for assets disposed of	Cash paid for assets purchased

Long-Term Debt	
	New Debt incurred for assets purchased

Depreciation Expense	
Depreciation expense for the current period	

Gain on Sale of Building (or Equipment)	
	Gain on sale
Loss on Sale of Building (or Equipment)	
Loss on sale	

Disposal of Non-Current Assets

Suppose you started the year with buildings that cost \$100,000. During the year, you bought another building for \$150,000 and ended the year with buildings that cost \$180,000. What was the cost of the building you sold?

Building			
Beg bal	100,000		
Cost of assets purchased	150,000	Cost of assets sold	X = 70,000*
End bal	180,000		

$$* X = 100,000 + 150,000 - 180,000$$

#5 Accounting for Specific Intangibles (1 of 5)

- **No physical form**
 - Carry special rights
 - Include patents, copyrights and franchises
- **Two categories**
 - **Finite lives**
 - Amortization recorded
 - Straight-line method
 - Intangible asset reduced directly
 - **Indefinite lives**
 - Tested for loss in value (impairment)

Accounting for Specific Intangibles

Patents

Copyrights

Trademarks
and trade
names

Franchises
& licenses

Goodwill

Assessment

Tutorial portfolio - Online quizzes:

This quiz is based on the week 4 learning materials. You may complete this anytime during week 4, you may use your study notes.

You have TWO attempts to complete this quiz.

Due time for your weekly quiz is on Canvas, please check it by yourself.